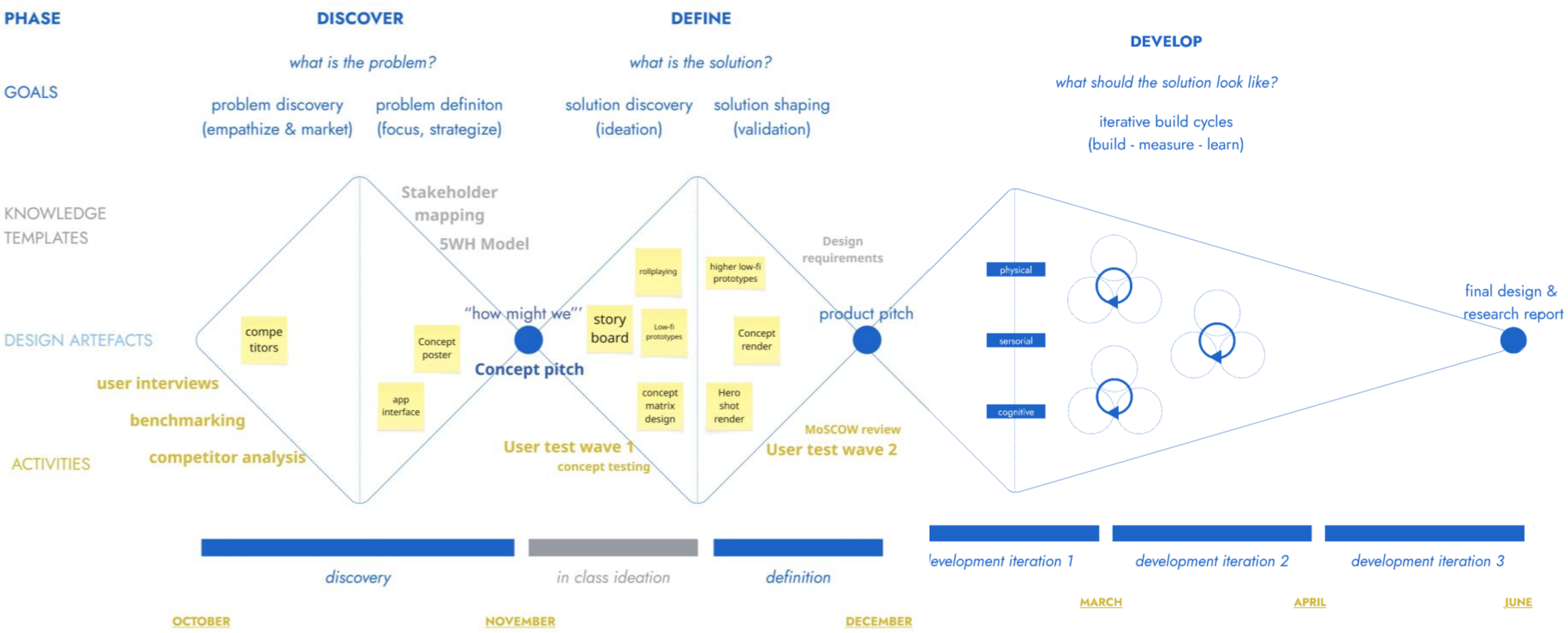
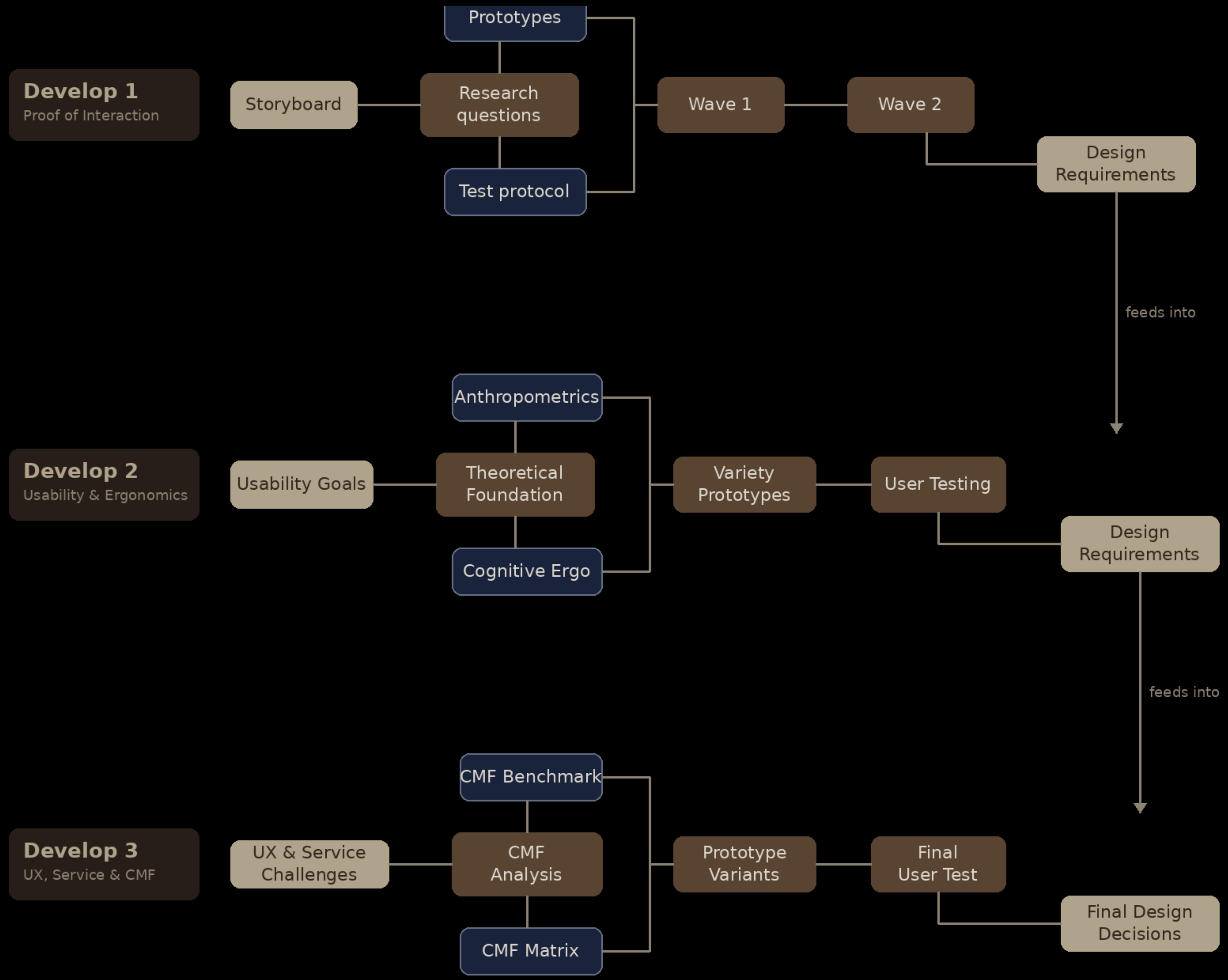
SensePath

Ear free, intuitive navigation in & outdoors


PHASE
DISCOVER
DEFINE
DEVELOP
what is the problem?
what is the solution?
what should the solution look like?
GOALS
problem discovery
(empathize & market)
problem definiton
(focus, strategize)
solution discovery
(ideation)
solution shaping
(validation)
iterative build cycles
(build - measure - learn)
KNOWLEDGE
TEMPLATES
Stakeholder
mapping
5WH Model
Design
requirements
DESIGN ARTEFACTS
compe
titors
Concept
poster
app
interface
"how might we"'
Concept pitch
story
board
rollplaying
Low-fi
prototypes
concept
matrix
design
higher low-fi
prototypes
Concept
render
Hero
shot
render
product pitch
physical
sensorial
cognitive
final design &
research report
ACTIVITIES
user interviews
benchmarking
competitor analysis
User test wave 1
concept testing
MoSCOW review
User test wave 2
discovery
in class ideation
definition
development iteration 1
development iteration 2
development iteration 3
OCTOBER
NOVEMBER
DECEMBER
MARCH
APRIL
JUNE

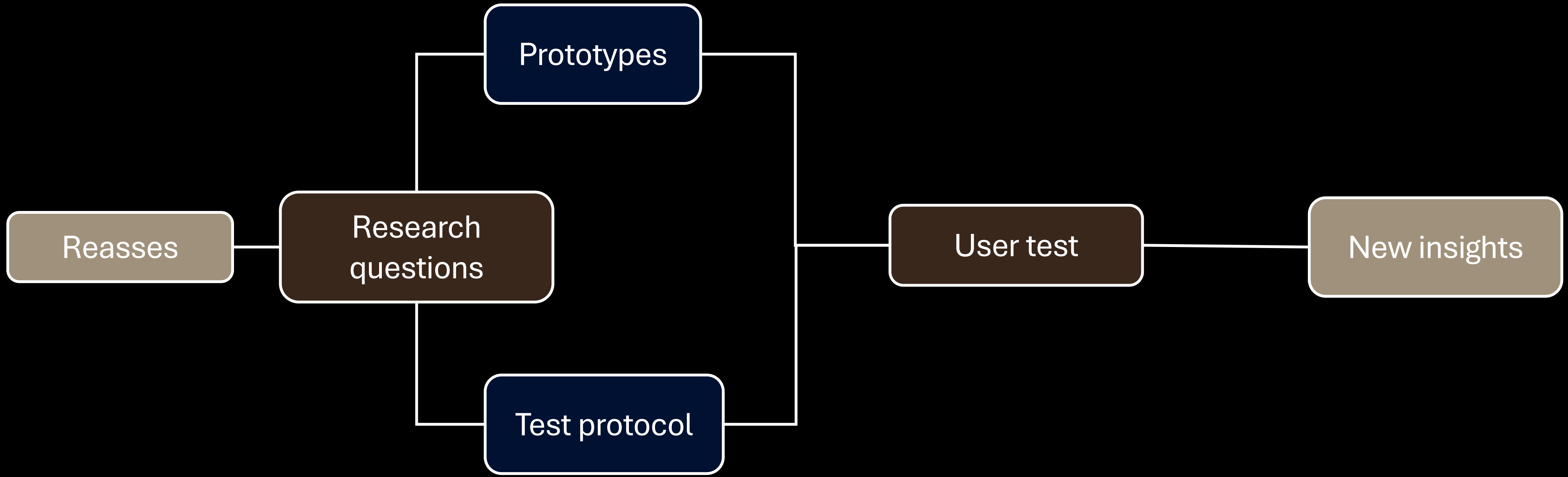
Develop

Methodology



Develop 3

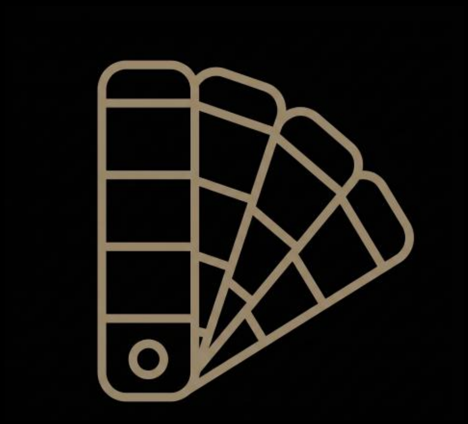
Methodology



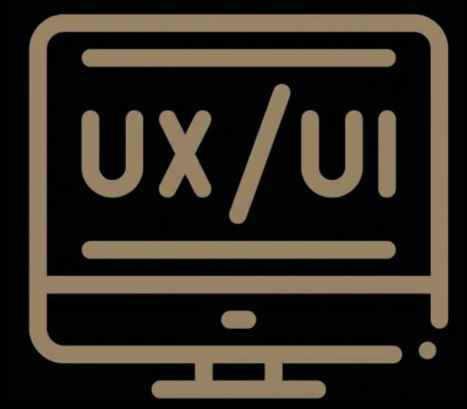
Research Questions

- Which set of **vibration patterns** (max. three to four signals via a single vibration motor) is sufficiently **distinguishable** and **intuitive** to communicate an obstacle, route deviation, and announcement of a turn without cognitive overload?
- At what **moment before a turn** does the user want to receive a preparatory vibration (expressed in steps, meters, or seconds)?
- Do users desire a single neutral directional **announcement signal** in combination with the compass, or an explicit distinction between left and right in the pattern?
- Which **handle material** (hard standard cane, softer compressible handle, cork element) is preferred in terms of tactility and usability in all weather conditions?
- Which **color choice** for the handle aligns with the need for contrast for the visually impaired and external visibility in public spaces?
- Does the perceived **cognitive load** and trust in a real-life context (traffic, pedestrians, guidelines, level differences) differ significantly from the lab setting in Develop 2?
- Do users desire **personal customizability** of vibration patterns and announcement distances, or does a fixed default setting suffice?

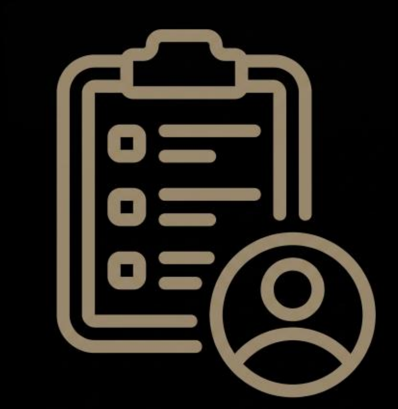
Goal



CMF Deepdive



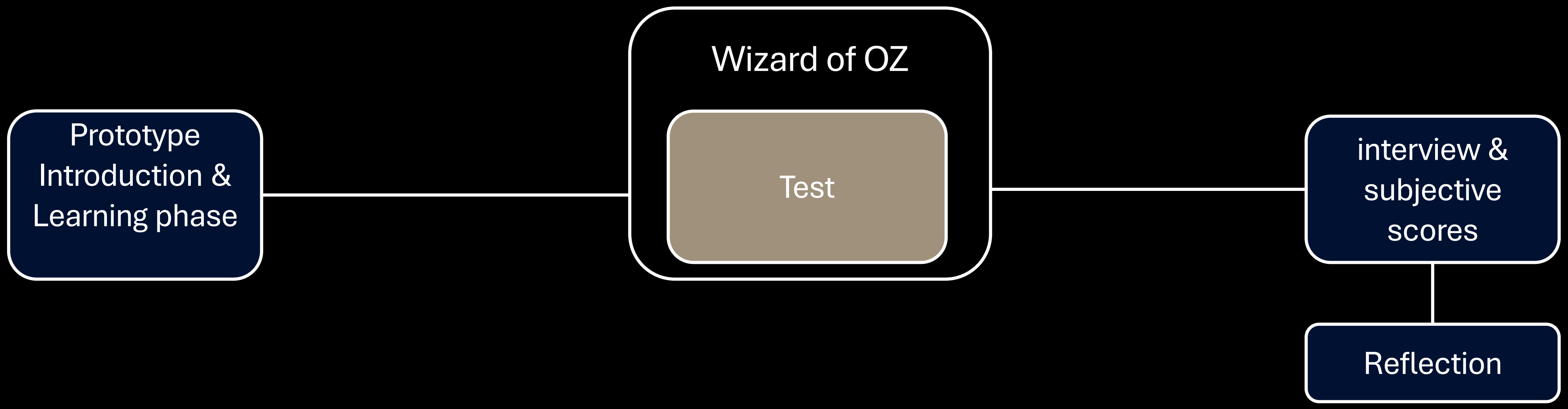
Assess **UX**



Final user test

UX test

Workflow



See protocol develop 3

Sample N=5

Name	Age	Relevance	Location	Time
Mario	52	Lost vision at age 30	Licht en liefde	07/05/2026
Jelle	22	Blind since birth	Licht en liefde	07/05/2026
Herman	65	Lost vision at age 18	Licht en liefde	08/05/2026
Milo	20	Sighted (control)	At home	08/05/2026
Milos	20	Sighted (control)	At home	08/05/2026

UX Feedback

- Patterns must remain **consistent**, even with multiple turns in quick succession
- Hesitated between announcement and obstacle pulse, **patterns are too similar to each other**
- **The feedback logic of the compass** (back to center = you are positioned correctly) was not immediately intuitive, but solvable through explanation
- Wants to be able to **recognize** the **direction** within the pattern itself
- Completely divergent feedback on turn announcements

"I still don't agree with the fact that you won't indicate which direction in your pulses.“

"Isn't that kind of the same thing?“

"It's really not that bad in terms of concentration."

"Rather indicating the direction where you already need to go."

Vibe check

Design implicaties

- Groove width is good, but onboarding needs to be clear
- Differentiate more sharply between announcement vs. obstacle in length or rhythm
- Direction as a vibration pattern still as the default setting (adjustable afterwards)
- Silent default for walking, no announcement for micro-corrections is good and clear
- Announcement distance is very personal and must be adjustable

- Softer handle material as the primary choice
- Color itself is not important, but contrast is -> black or red handle
- The white cane itself is required to be white
- Replaceable compass pin is necessary

CMF

Handle

Handle

Handle

Component	Function	Why CMF-relevant
Structural core	Carries electronics (PCB, battery, LRA motor); provides threading for mounting	Lost vision at age 18
Soft-touch overmold	55	Severely visually impaired
Grip texture	Increases grip, helps user orient handle in hand	Pitch and depth of ribs/dots determine tactile legibility
Button	On/off, Smart-feature control	Must be tactilely distinct from other buttons via relief or contrast
Pin tip / pointer head	The end the user feels with the thumb to read direction	Most touched single point on the entire product — tactile shape and material are critical

Structural core

Criterion (weight)	PA6-GF30	PA6 unfilled	Polycarbonate (PC)
Rigidity / dimensional stability (×3)	5	3	4
TPE overmold adhesion (×3) — *Persson 2020*	1	5	5
Weight (×2)	4	4	3
UV resistance (×1)	4	4	2
Production cost (×2)	4	5	3
2K line availability BE/NL (×1)	5	5	4
Total (max 60)	**40**	**49**	**44**

Structural core

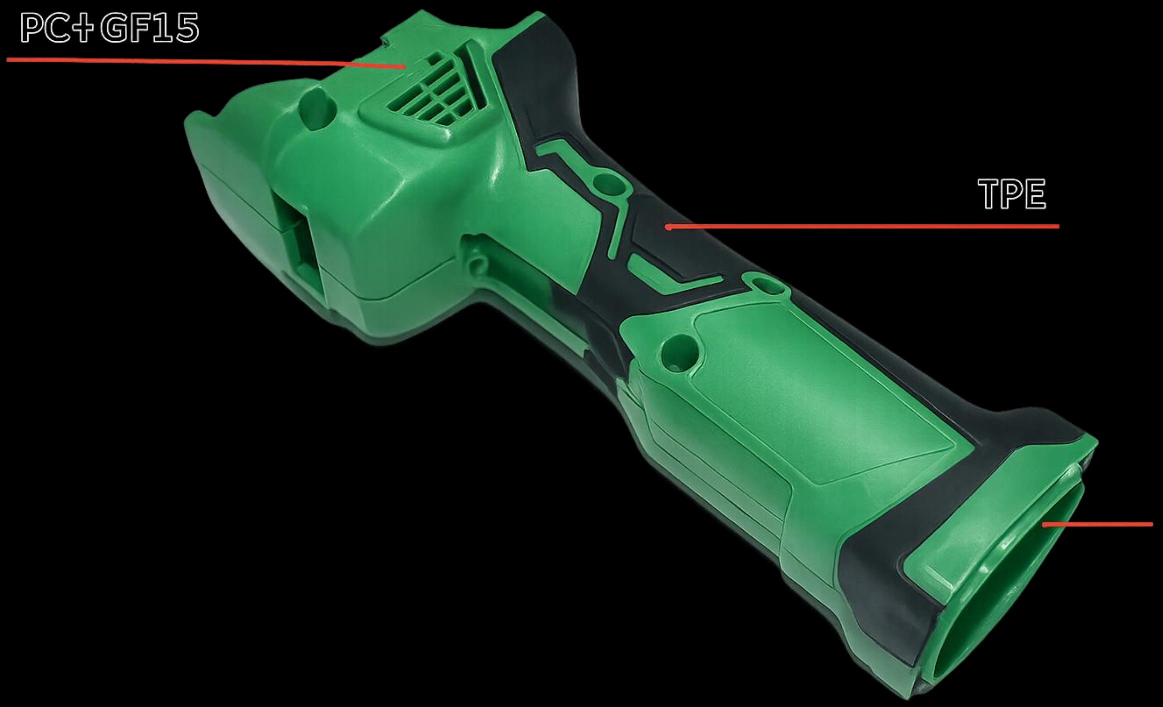
PA6 unfilled example

Soft-touch overmold (grip)

Criterion (weight)	TPE Shore 65A	TPU Shore 70A	Silicone Shore 70A
Tactile comfort over 8h use (×3)	5	4	4
Grip in wet conditions (×3)	4	5	4
2K overmold compatibility (×3)	5	4	2
UV and colour fastness (×2)	4	4	3
Recyclability (×1)	4	3	1
Production cost (×2)	5	3	2
Wear resistance (×1)	3	5	4
Total (max 75)	**62**	**57**	**42**

Soft-touch overmold

TPE Shore 65A example



Grip texture (mould-engraved pattern)

Criterion (weight)	Fine radial ribs (pitch 0.8 mm)	Diamond pattern (pitch 1.0 mm)	Linear ribs (pitch 1.2 mm)
Above tactile threshold for blind users (×3) — *Alary 2008*	5	5	5
Age-resilient legibility (×3) — *Skedung 2018*	5	5	5
Grip in wet conditions (×2)	5	4	4
Cleanability / dirt traps (×2)	4	2	4
Visual restraint (Quiet aesthetic) (×2)	5	3	4
Mould engraving cost (VDI/Mold-Tech) (×1)	4	3	5
Hand orientation cue (×1)	5	3	4
Total (max 70)	**63**	**48**	**57**

Grip texture

Fine radial ribs (pitch 0.8 mm) example

Buttons (cap material)

Criterion (weight)	POM (Delrin) + TPE 50A overlay	ABS + chrome ring	Silicone 60A solid
Click feel after 10,000 cycles (×3)	5	4	3
Tactile distinction from grip (×3) — *Norman 2011*	5	3	4
Laser-engravable for icons (×2)	4	5	2
IPX5 sealing compatibility (×2)	5	4	5
Production cost (×1)	3	5	4
Visual integration with grip (×2)	5	3	4
Total (max 65)	**57**	**48**	**44**

Buttons

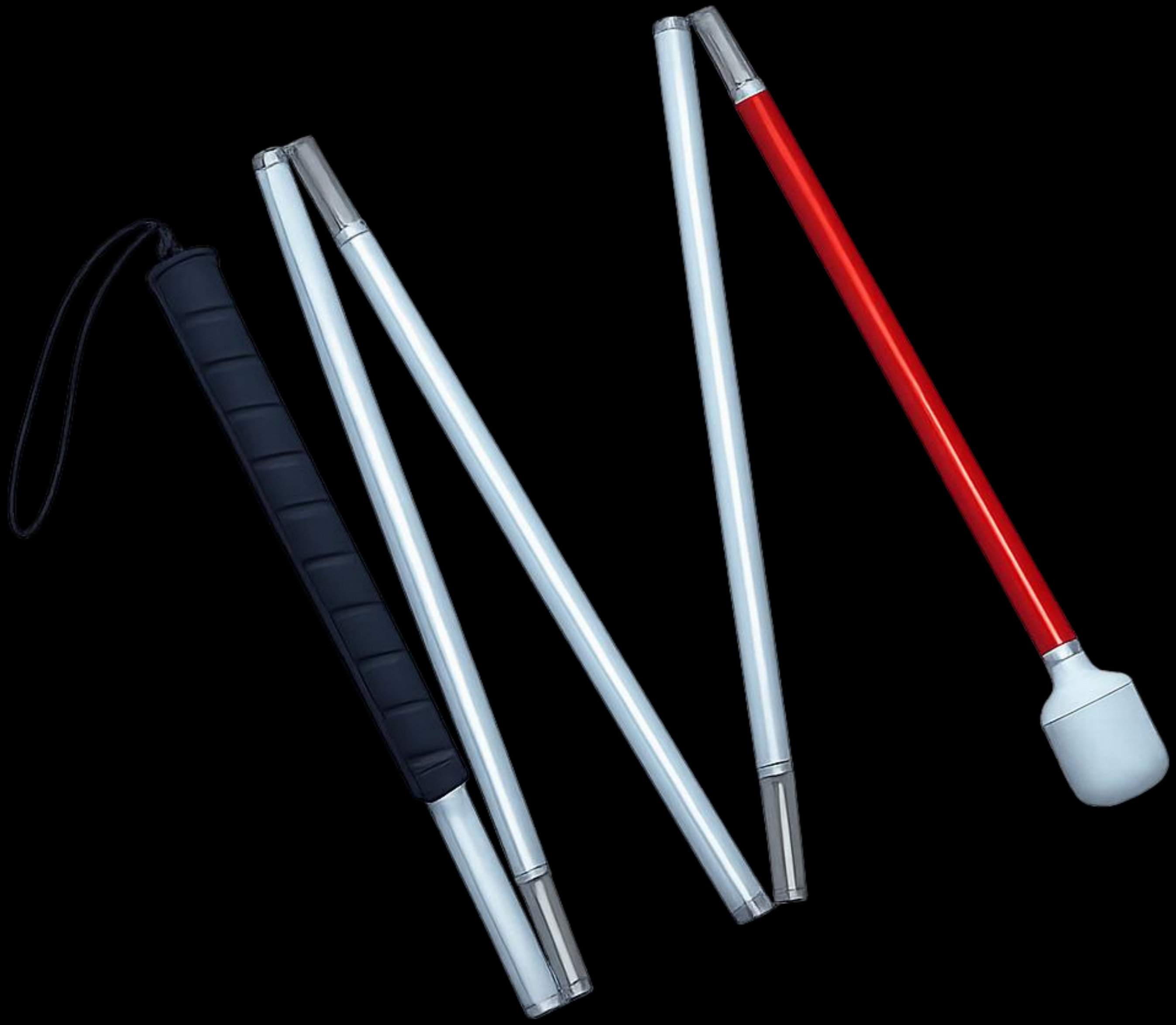
POM (Delrin) + TPE 50A overlay example

Pin tip (compass indicator)

Criterion (weight)	TPE 50A on aluminium core	POM (Delrin)	Anodised aluminium 6061
Thermal comfort (cold morning, wet thumb) (×3)	5	4	1
Tactile readability of direction notch (×3)	5	4	5
Wear resistance (rotation, thumb friction) (×3)	4	4	5
Premium feel (×2)	5	2	5
Low-friction rotation (×2)	4	5	4
Production cost (×1)	3	5	3
Weight at the front of the handle (×1)	4	5	3
Total (max 75)	**64**	**53**	**53**

Cane

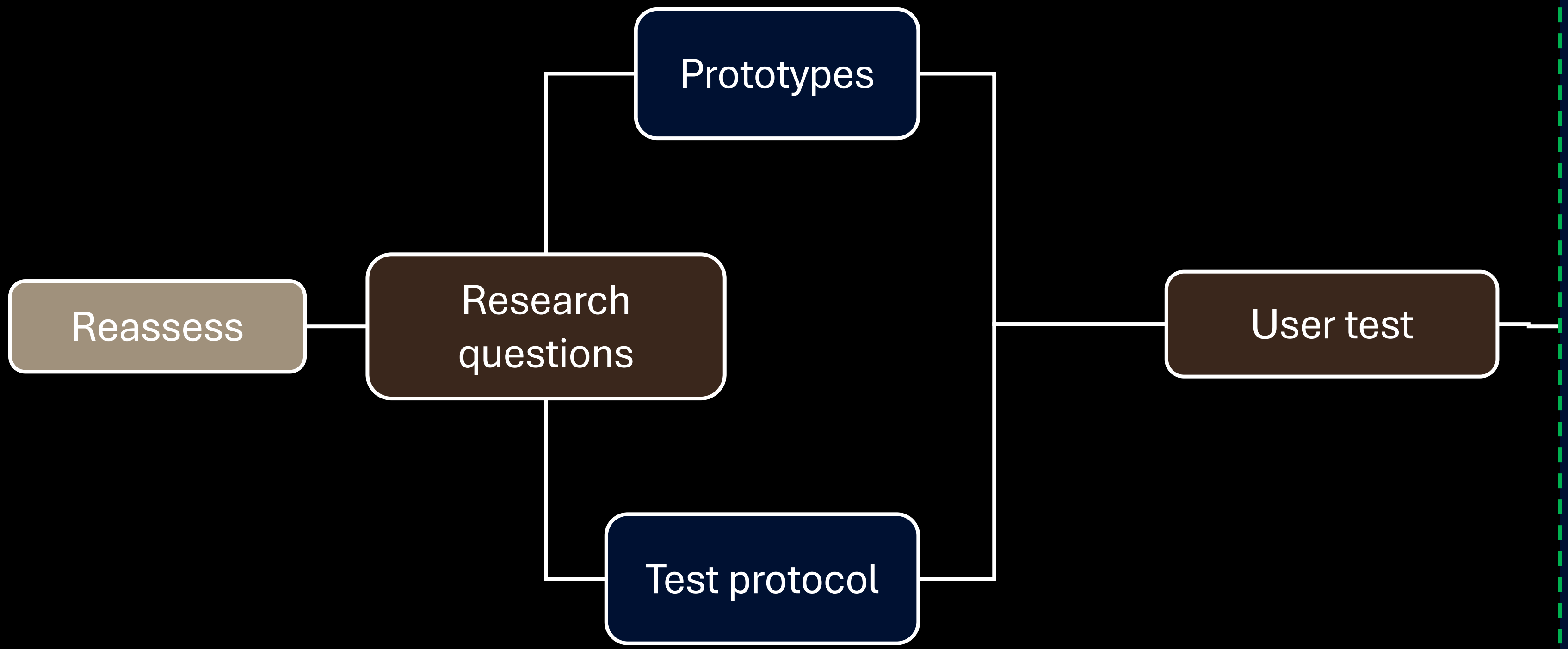
Cane



Cane *ISO* 9999

Component	Function	Why CMF-relevant
Shaft outer diameter	~12 mm	WHO APS24
Shaft material	Aluminium, fibreglass, or graphite/carbon composite	WHO APS24, Wikipedia
Length	Based on user height (— typically 48–60 inch / 122–152 cm	Leader Dogs, BAWA
Tip material	Nylon or TPU	WHO APS24
Tip diameter	2–3 cm	WHO APS24
Colour coding	White (upper) + red (lower) — legally required in most countries	National traffic laws
Expected lifespan	Min. 2 years under normal use	WHO APS24
Folding mechanism	Internal elastic cord (latex/polyester) running through all segments	WHO APS24

Develop 1 conclusions



- Direction must be encoded in the vibration pattern itself.
- Announcement and obstacle pulses need sharper rhythmic differentiation.
- Announcement distance: customizable. Vibration patterns: fixed defaults.
- Real-world cognitive load is lower than lab predicted.
- Softer handle, high contrast, white cane, replaceable pin.

References: Academic literature

- Persson, F., Bergström, G., Hagström, B., & Boldizar, A. (2020). Adhesion between thermoplastic elastomers and polyamide-12 with different glass fiber fractions in two-component injection molding. *Polymer Engineering & Science*, 60(8), 1858–1869. https://doi.org/10.1002/pen.25408
- Norman, J. F., & Bartholomew, A. N. (2011). Blindness enhances tactile acuity and haptic 3-D shape discrimination. *Attention, Perception, & Psychophysics*, 73(7), 2323–2331. https://doi.org/10.3758/s13414-011-0160-4
- Skedung, L., El Rawadi, C., Arvidsson, M., Farcet, C., Luengo, G. S., Breton, L., & Rutland, M. W. (2018). Mechanisms of tactile sensory deterioration amongst the elderly. *Scientific Reports*, 8, 5303. https://doi.org/10.1038/s41598-018-23688-6
- Alary, F., Duquette, M., Goldstein, R., Elaine Chapman, C., Voss, P., La Buissonnière-Ariza, V., & Lepore, F. (2009). Tactile acuity in the blind: A closer look reveals superiority over the sighted in some but not all cutaneous tasks. *Neuropsychologia*, 47(10), 2037–2043.
- Arthur, P., & Passini, R. (1992). *Wayfinding: People, signs, and architecture*. McGraw-Hill.
- Liu, G., Yu, T., Yu, C., Xu, H., Xu, S., Yang, C., Wang, F., Mi, H., & Shi, Y. (2021). Tactile Compass: Enabling visually impaired people to follow a path with continuous directional feedback. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM. https://doi.org/10.1145/3411764.3445644

References: Industrial and technical sources

- Kraiburg TPE. (2023). *THERMOLAST series — TPEs for tool handles combine performance with sustainability*. Retrieved from https://www.kraiburg-tpe.com/en/tpe-tool-handles-combine-performance-sustainability
- Teknor Apex. *Monprene thermoplastic elastomers product guide* (CP-10100 / CP-11100 series). Retrieved from https://www.teknorapex.com/en-us/resources/technical-literature/monprene-thermoplastic-elastomers-product-guide
- Teknor Apex. *Shore Hardness Chart for TPEs — Shore A vs Shore D*. Retrieved from https://www.teknorapex.com/en-us/resources/blogs/thermoplastic-elastomers-and-measuring-shore-hardness-chart
- Boyan Manufacturing Solutions. (2023). *TPE overmolding of a power tool handle on PA6+GF30 substrate*. Retrieved from https://boyanmfg.com/shop/tpe-overmolding-power-tool-handle/
- Franplast. *Chemiton 2K — TPE for overmolding manufacturers (VDI 2019 adhesion testing)*. Retrieved from https://www.franplast-tpe.com/tpe-for-overmolding-manufacturers/
- Plasticmold.net. *What is overmolding and 2K injection molding*. Retrieved from https://www.plasticmold.net/what-is-overmolding-and-2k-injection-molding/
- 3M. *Diamond Grade DG3 Reflective Sheeting — datasheet*.
- Reflexite / ORALITE. *DOT-C2 Reflective Conspicuity Tape — V92308 datasheet*.
- Ambutech. *Cane components and tip systems — product documentation* (Slip-On / Hook).

References: Standards

- ISO 9999:2016. *Assistive products: Classification and terminology* — code 12 39 03 Tactile sticks or white canes. International Organization for Standardization.
- ISO 23599:2019. *Assistive products for blind and vision-impaired persons — Tactile walking surface indicators*. International Organization for Standardization.
- EN 12182. *General requirements for assistive products for persons with disabilities* (CE-marking). European Committee for Standardization.
- ISO 10993-1. *Biological evaluation of medical devices — Part 1: Evaluation and testing within a risk management process*.
- IEC 60529. *Degrees of protection provided by enclosures (IP Code)*.
- VDI 2019. *Adhesion testing for two-component injection moulding*. Verein Deutscher Ingenieure.
- ASTM D 4956. *Standard Specification for Retroreflective Sheeting for Traffic Control* (Type V — DOT-C2 / DOT-C3 / DOT-C4).
- World Health Organization. *Assistive Product Specification APS24 — White canes procurement guidelines* (based on ISO 9999).

References: Course material

- UGent. *Industriële Vormgeving Les 3 — Colors, Materials & Finishes (CMF)*. Universiteit Gent, Industrial Design Engineering.

References: AI-disclosure

•**Anthropic. (2025).** Claude (Claude Opus 4.6) [Groot taalmodel]. https://claude.ai
• **OpenAI. (2025).** ChatGPT (GPT-4o) [Groot taalmodel]. https://chat.openai.com
• **Google DeepMind. (2025). Gemini** [Groot taalmodel]. https://gemini.google.com
• **Google DeepMind. (2025). Nano Banana Pro** (Gemini 3 Pro Image) [AI-beeldgeneratiemodel]. https://blog.google/innovation-and-ai/products/nano-banana-pro/

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